Bags, Vectors & Transformers

Bias, Measurement & What Comes Next · Day 2, Part 3

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Welcome Back

You have now worked with real embeddings: nearest neighbors, analogies, your own Word2Vec model. This part steps back to ask the questions that matter most for **research**:

- ⚠ What do embeddings absorb from their training data — and why should we worry?
- 📈 How can that same property be turned into a **measurement tool** for social science?
- ⚖ How do we use embeddings **responsibly** — validation and pitfalls?
- → Where do static embeddings fall short, and what comes next?

Did anyone's analogy in the notebook return something odd? Let's start there.

Embeddings Absorb Bias

You may have seen it yourself in the notebook: some analogies return **stereotyped** results.

man − woman ≈ computer programmer − homemaker

(Bolukbasi et al., 2016)

This is not a bug in the algorithm. Embeddings faithfully learn the **statistical patterns** in their training text — and human text is full of **social bias**. The geometry encodes it: gender, race, age, and more.

If you use off-the-shelf embeddings in a pipeline, these biases ride along silently.

Why Bias Matters — Two Faces

As a hazard

· Bias propagates into **downstream models**

· A CV-screener or classifier can **inherit discrimination**

· Often **invisible** unless you test for it

· An ethical and validity problem

As a measurement

· The bias in the vectors **reflects the bias in society**

· So we can **measure** stereotypes by measuring the geometry

· Track how attitudes **change over time**

· A research **opportunity**

Same fact, two uses. The rest of this section leans into the second — embeddings as a lens on society.

Embeddings as Measurement

A powerful idea for social science: if meaning is geometry, then **social meaning** is geometry too. We can **quantify** associations that would be hard to measure otherwise.

The recipe: take a concept (say, a profession) and measure its **distance** to two poles (say, "he/him" vs "she/her"). The relative distance is a **numeric measure of association** — a stereotype score you can compare across groups, corpora, or decades.

Example studies:

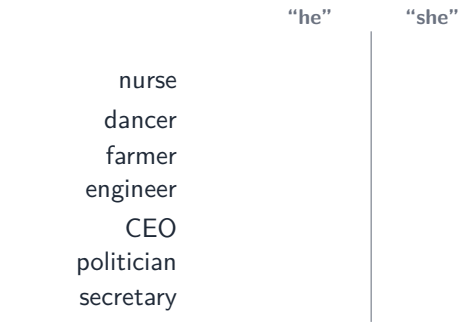
- Garg et al. (2018): 100 years of gender & ethnic stereotypes
- Kozlowski et al. (2019): "the geometry of culture" — class, gender, race

What you can study:

- Framing of groups in the news
- Change in public discourse over time
- Cross-national or cross-corpus comparison

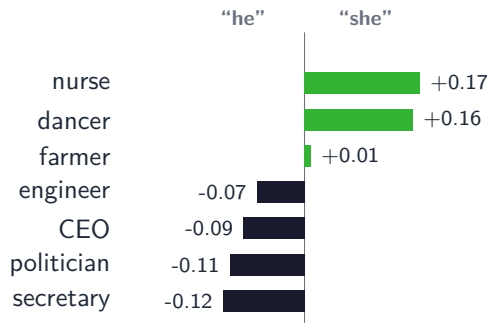
She or He?

YOUR TURN Hands up: in GloVe, is each occupation closer to "**she**" or to "**he**"?



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Score: cosine(word, she) − cosine(word, he), GloVe 50d.

Mostly the stereotypes you would expect. But "secretary" leans **he**: in news it is mostly "Secretary of State" or "general secretary".

One word, several senses, one vector. Use **sets** of words for each pole, and check what drives a score.

Measuring Change Over Time

The trick that makes this powerful for social science: train (or obtain) **separate embeddings for different time periods**, then track how a word's associations **move**.

embed each period → measure "woman" near "career" vs "home" → plot the trend



Suddenly a fuzzy question — "how have gender roles in public language shifted since 1910?" — becomes a **measurable time series**. This is embeddings doing genuinely novel social-science work.

Validation & Pitfalls

Embeddings are seductive — the numbers feel objective. But they come with real traps. Treat them with the same skepticism as any measurement instrument.

Common pitfalls

- **Instability**: different runs give different vectors
- **Corpus size**: small corpora give noisy embeddings
- **Over-reading** 2-D plots
- Treating a single number as ground truth

Good practice

- Train **several models**, check stability
- Report **uncertainty**, not just point estimates
- **Validate** against human judgment or known cases
- Read some actual **text** — don't trust geometry blindly

The computational result is a starting point for interpretation, not the end of it.

Convince a Skeptical Reviewer

TURN TO YOUR NEIGHBOR 3 minutes

You score how **populist** each party's manifesto is, by measuring how close its words are to a "populism" direction in embedding space. A reviewer asks: *how do you know this measures populism?* What would you show them?

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Some answers

- **Known cases**: do parties everyone agrees are populist score high?
- **External benchmarks**: correlate with expert surveys of parties (e.g., CHES, POPPA)
- **Human coding**: hand-code a sample of passages and compare
- **Read** the highest- and lowest-scoring passages
- **Robustness**: other word lists, other embeddings, several training runs

Guess the Hidden Word: One Word, Three Meanings

YOUR TURN One word, hidden in all three sentences. What is it?

1 The _____ needs more water and light.

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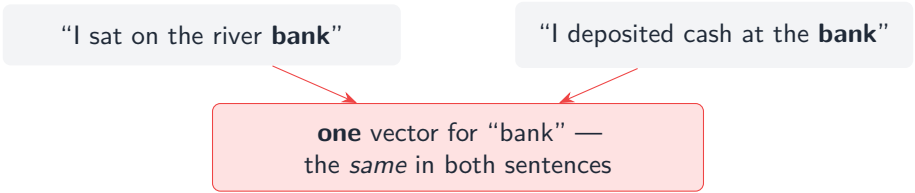
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plant You switched meanings effortlessly. A static embedding can't: it has **one** vector for "plant". GloVe's neighbors: *plants, factory, facility, produce, farm, chemical, fertilizer*, a blend leaning toward factories. Not great news for plant scientists in Wageningen.

The Limit of Static Embeddings

Static embeddings were a huge leap — but they share one stubborn flaw with bag-of-words. Remember "**bank**"?



A static embedding gives every word **exactly one vector**, no matter the sentence. So the two very different "banks" collapse into one. **Context is still being ignored** — just at the word level instead of the document level.

Where Did the River Go?

YOUR TURN GloVe has **one** vector for "bank". Which words do you expect among its nearest neighbors?

bank

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banks securities banking investment exchange financial

No river in sight. The single vector is dominated by the most frequent sense in the training data. Rarer senses get averaged away: exactly the problem contextual embeddings solve.

The Fix: Contextual Embeddings

Contextual embeddings (BERT and its relatives) solve this: instead of one fixed vector per word, they compute a **fresh vector for each word every time it appears**, based on its surrounding sentence.

"river **bank**" → vector A	"cash at the **bank**" → vector B

vector A ≠ vector B — context decides the meaning

The two "banks" now get **different vectors**. The model reads the whole sentence and shapes each word's representation to fit. This is the leap from **static** to **contextual**.

How? A One-Slide Peek at Attention

The engine behind contextual models is the **transformer**, and its key mechanism is **attention**. The intuition is simple:

> When building the representation of a word, the model **looks at every other word in the sentence** and decides **how much each one matters** for understanding it. For "bank", the word "river" or "cash" gets high attention — and that is what pins down the meaning.

That is deliberately simplified — transformers are a course of their own. What matters for us today: **attention is how context gets baked into every word's vector**.

We will work with these models hands-on later in the course.

Part 2: The Takeaways

- ✔ Embeddings **absorb social bias** from their training data — a hazard, but also a **measurement opportunity**
- ✔ Embeddings-as-measurement lets us **quantify meaning** and track it **over time** — genuinely novel social science
- ✔ They demand **careful validation**: check stability, report uncertainty, read the text
- ✔ **Static** embeddings still give one vector per word — context-blind at the word level
- ✔ **Contextual** embeddings (BERT, via attention) give a vector *per use* — the next leap

That completes our tour from **bags** to **vectors** — next, the **transformers**.